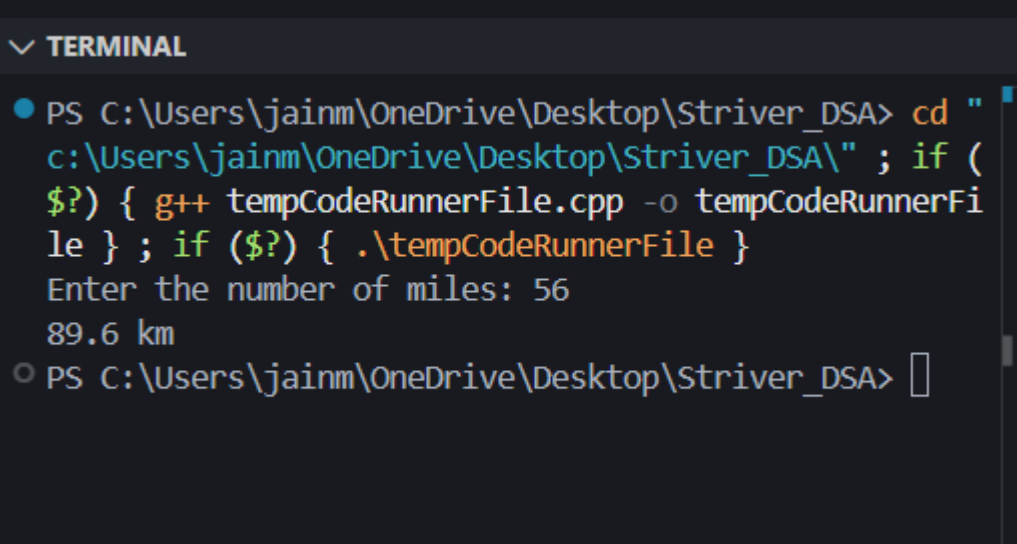


Submitted by: Madhav Jain 1024040079

Q1: Write a program in C to convert miles into kilometers (Km). Hint: 1 Mile=1.609 Km.
[Use macros, relevant name and types for variables].

```
#include<iostream>
using namespace std;
#define miles_to_kms 1.6
int main(){
    double miles, kms;
    cout<<"Enter the number of miles: ";
    cin>>miles;
    kms=miles*miles_to_kms;
    cout<<kms<<" km";
    return 0;
}
```



The screenshot shows a terminal window with a dark background. At the top, there is a tab labeled 'TERMINAL'. The terminal displays the following commands and output:

```
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
Enter the number of miles: 56
89.6 km
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> 
```

Q2: Write a program to find the number of positive, negative and zeros in a sequence of inputs (numbers) entered as data.

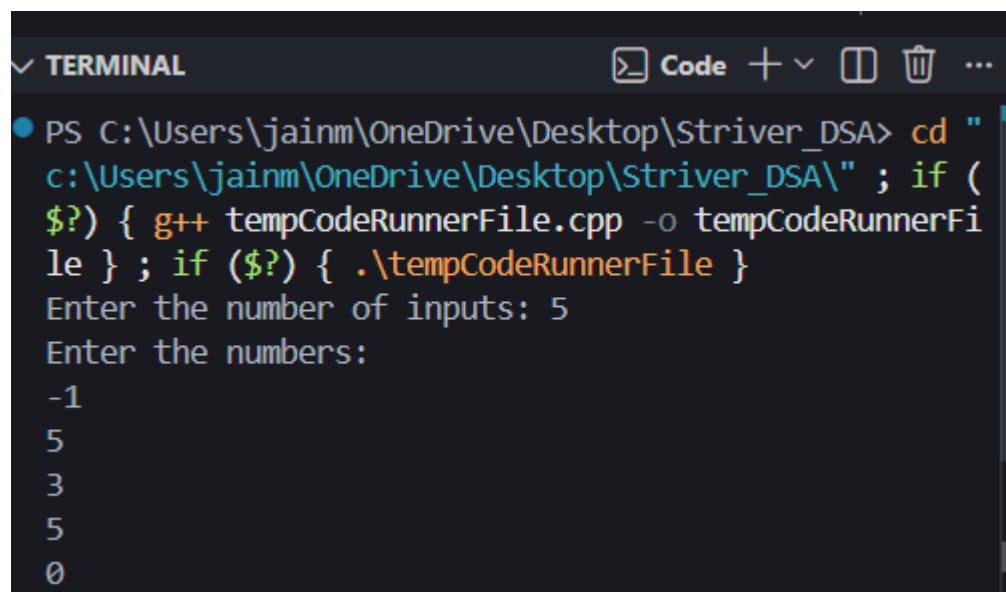
```
#include <iostream>
using namespace std;

int main() {
    int n,num;
    int positive=0,negative=0,zero=0;
    cout<<"Enter the number of inputs: ";
    cin>>n;
    cout<<"Enter the numbers:"<<endl;
    for(int i=0; i<n; i++){
        cin>>num;
```

```

    if(num>0)
        positive++;
    else if (num<0)
        negative++;
    else
        zero++;
}
cout<<"Positive numbers: " << positive<<endl;
cout<<"Negative numbers: " << negative<<endl;
cout<<"Zeros: " << zero<<endl;
return 0;
}

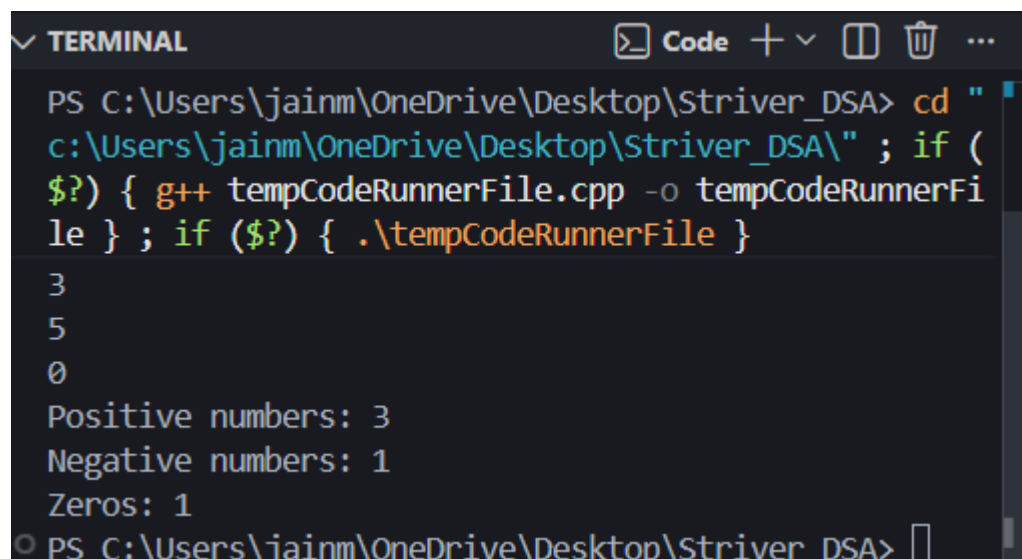
```



```

✓ TERMINAL
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
Enter the number of inputs: 5
Enter the numbers:
-1
5
3
5
0

```



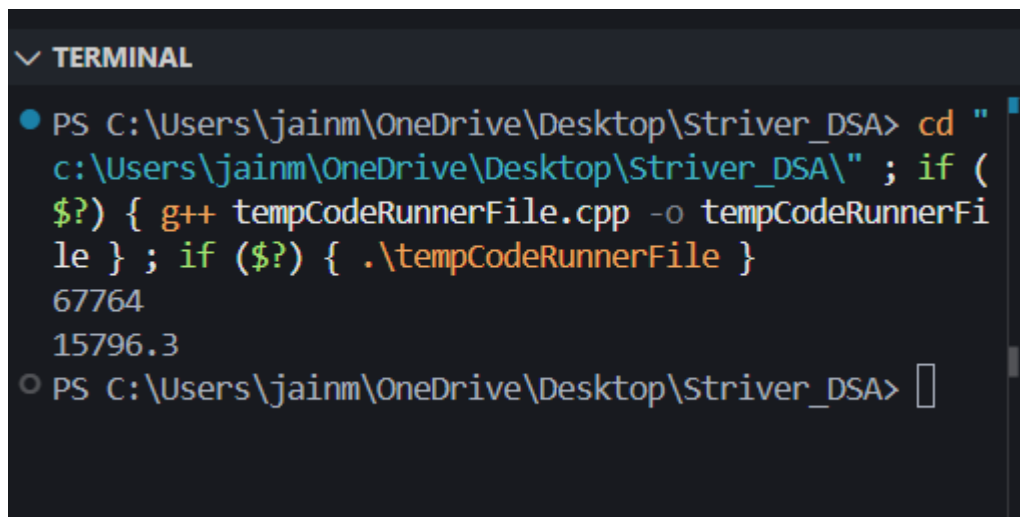
```

✓ TERMINAL
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
3
5
0
Positive numbers: 3
Negative numbers: 1
Zeros: 1
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA>

```

Q3: Compute the tax due based on a tax table given below: Program Input: Salary amount.
Program Output: Returns the tax due for $0.0 \leq \text{salary} \leq 150,000.00$; returns -1.0 if salary is outside the table range.

```
#include<iostream>
using namespace std;
int main(){
double salary,tax;
cin>>salary;
if(salary<0||salary>150000)
tax=-1.0;
else if(salary<=14999.99)
tax=salary*0.15;
else if(salary<=29999.99)
tax=2250+(salary-15000)*0.18;
else if(salary<=49999.99)
tax=5400+(salary-30000)*0.22;
else if(salary<=79999.99)
tax=11000+(salary-50000)*0.27;
else
tax=21600+(salary-80000)*0.33;
cout<<tax;
return 0;
}
```



```
✓ TERMINAL
● PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
67764
15796.3
○ PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> 
```

Q4: Write an interactive program (menu driven) in 'C' (using functions) to compute the area of a selected geometrical figure from a list of such figures (square, rectangle, and circle).

```
#include<iostream>
using namespace std;
```

```
double square(double side){
return side*side;
}
double rectangle(double length,double width){
return length*width;
}
double circle(double radius){
return 3.14159*radius*radius;
}
```

```
int main(){
int choice;
double a,b;
cout<<"1. Square\n2. Rectangle\n3. Circle\n";
cout<<"Enter your choice: ";
cin>>choice;
```

```
switch(choice){
case 1:
cout<<"Enter side: ";
cin>>a;
cout<<"Area = "<<square(a);
break;
case 2:
cout<<"Enter length and width: ";
cin>>a>>b;
cout<<"Area = "<<rectangle(a,b);
break;
case 3:
cout<<"Enter radius: ";
cin>>a;
cout<<"Area = "<<circle(a);
break;
default:
cout<<"Invalid choice";
}
return 0;
}
```

```
✓ TERMINAL
● PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
1. Square
2. Rectangle
3. Circle
Enter your choice: 3
Enter radius: 5
Area = 78.5397
○ PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> 
```

Q5: Write a program to display first n elements of Fibonacci series.

```
#include<iostream>
using namespace std;
int main(){
int n,a=0,b=1,c;
cin>>n;
for(int i=0;i<n;i++){
cout<<a<<" ";
c=a+b;
a=b;
b=c;
}
return 0;
}
```

```
✓ TERMINAL
● PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
5
0 1 1 2 3
○ PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> 
```

Q6: Write a program to print a table book from Table X to Table Y. X and Y are user inputs.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int X, Y, i, j;
```

```
    printf("Enter X: ");
```

```
    scanf("%d", &X);
```

```
    printf("Enter Y: ");
```

```
    scanf("%d", &Y);
```

```
    for (i = X; i <= Y; i++)
```

```
    {
```

```
        printf("\nTable of %d:\n", i);
```

```
        for (j = 1; j <= 10; j++)
```

```
        {
```

```
            printf("%d x %d = %d\n", i, j, i * j);
```

```
        }
```

```
    }
```

```
    return 0;
```

```
}
```

Terminal

Code + ▾ □ □ ...

```
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
```

Enter X: 2

Enter Y: 4

Table of 2:

2 x 1 = 2

2 x 2 = 4

2 x 3 = 6

2 x 4 = 8

2 x 5 = 10

2 x 6 = 12

2 x 7 = 14

2 x 8 = 16

2 x 9 = 18

2 x 10 = 20

Table of 3:

3 x 1 = 3

3 x 2 = 6

3 x 3 = 9

3 x 4 = 12

3 x 5 = 15

3 x 6 = 18

3 x 7 = 21

3 x 8 = 24

3 x 9 = 27

3 x 10 = 30

Terminal

Code + ▾ □ □ ☒ ...

```
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
```

```
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20
```

Table of 3:

```
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
```

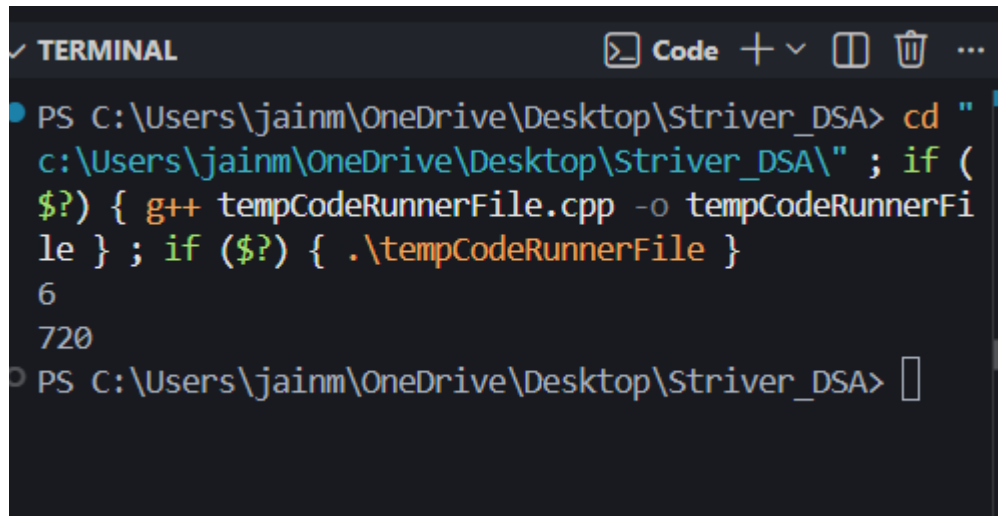
Table of 4:

```
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40
```

```
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> 
```


Q7: Write a program to compute factorial of a number using iterative approach.

```
#include<iostream>
using namespace std;
int main(){
int n,fact=1;
cin>>n;
for(int i=1;i<=n;i++)
fact*=i;
cout<<fact;
return 0;
}
```

A screenshot of a terminal window with a dark background. The title bar says 'TERMINAL'. The command prompt shows the user is in the directory 'C:\Users\jainm\OneDrive\Desktop\Striver_DSA'. They run a command to compile and run a C++ program: 'cd "c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (\$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if (\$?) { .\tempCodeRunnerFile }'. The program takes the input '6' and outputs '720'.

```
✓ TERMINAL Code + - [ ] [ ] [ ] ...
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
6
720
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> [ ]
```

Q8: Write a program to swap two numbers using functions.

```
#include<iostream>
using namespace std;

void swap(int &a,int &b){
int temp=a;
a=b;
b=temp;
}
```

```
int main(){
int a,b;
cin>>a>>b;
swap(a,b);
cout<<a<<" "<<b;
return 0;
}
```

```
✓ TERMINAL
● PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
5
6
6 5
○ PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> 
```

Q9: Write a function that returns the first integer between n_min and n_max entered as data to the calling function (main).

```
#include<iostream>
using namespace std;
```

```
int findNumber(int n_min,int n_max){
int n;
while(cin>>n){
if(n>=n_min&& n<=n_max)
return n;
}
return -1;
}
```

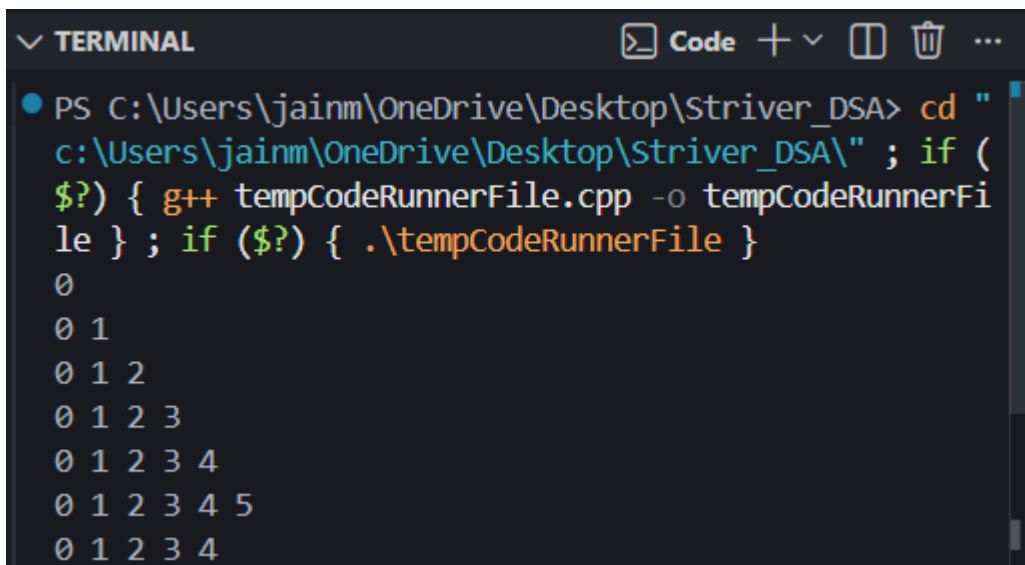
```
int main(){
int n_min,n_max,result;
cin>>n_min>>n_max;
result=findNumber(n_min,n_max);
cout<<result;
return 0;
}
```

```
✓ TERMINAL Code + ▾ □ □ □ ...
● PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
67
646
87
87
○ PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> □
```

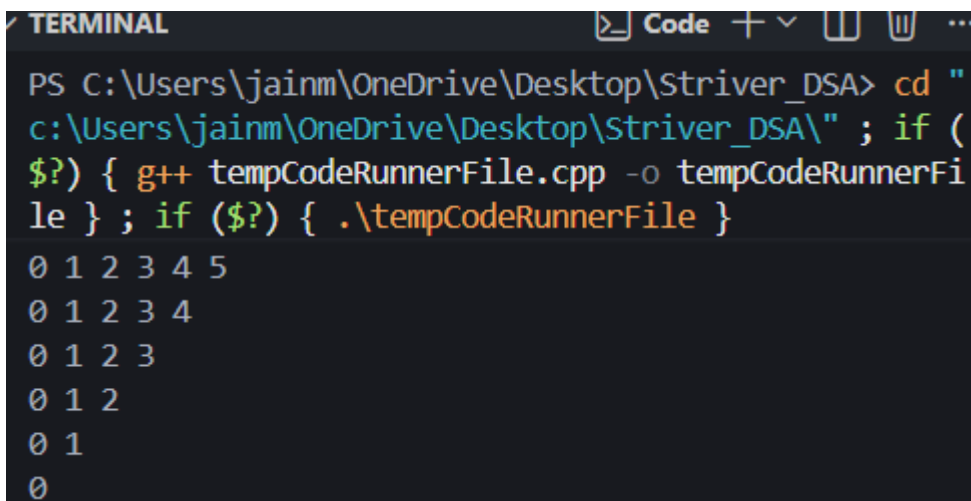
Q10: Write nests of loops that cause the following output to be displayed.

```
#include<iostream>
using namespace std;
```

```
int main(){
for(int i=0;i<=5;i++){
for(int j=0;j<=i;j++)
cout<<j<<" ";
cout<<endl;
}
for(int i=4;i>=0;i--){
for(int j=0;j<=i;j++)
cout<<j<<" ";
cout<<endl;
}
return 0;
}
```



```
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
0
0 1
0 1 2
0 1 2 3
0 1 2 3 4
0 1 2 3 4 5
0 1 2 3 4
```



```
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "
c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if (
$?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFi
le } ; if ($?) { .\tempCodeRunnerFile }
0 1 2 3 4 5
0 1 2 3 4
0 1 2 3
0 1 2
0 1
0
```